

ActInf Textbook Group ~ Cohort 3 ~ Meeting 23 (Chapter 10, part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_3 | Cohort 3 ~ Meeting 23 (Chapter 10, part 2) | 1:09:31

All right. Greetings, everyone. Thanks for joining. It is August 15th, and we're in our final discussion, recorded discussion for Cohort 3.

Next week, we'll have a kind of unrecorded discussion. Also, this will be the day of the symposium, so a very auspicious day to have been concluded upon.

But today, we're going to talk about 10 and probably anything else.

So, Ali or Neil, want to leave with anything? And or we can go to Esmails' question first.

Right. I've quickly put in the last three questions, but I think we can get through those.

So, we can start with Esmails.

All right. I'll just read it. And then happy to hear anyone's thoughts or more details.

Since I believe in Occam's razor, I deny the meaning of awareness.

My suggestion is accessible cognition and inaccessible cognition.

It looks like we have an operating system and implicit memory that's not available, such as the procedural memory and emotional memory.

I have some doubts about the rest of the words like consciousness and awareness.

And this is about the question, the opening quotation of the chapter.

Beginning of the end.

In general, we are at least aware of what our minds do best.

Neil or Ali?

That touches on my last question, which was what is sentient about Active Inference.

That's put in the chapter title.

Well, I've suggested as a teaser.

The chapter's really just summarising.

uh it's it doesn't doesn't really say anything about consciousness I don't think

and where Carla's it all seems to be getting more involved in discussions of consciousness but I

think he always maintains a distance between that and and works with with uh people who are concerned

with that but it's it's more something where active inferences is a framework for um well to help

people who are looking at sentence and consciousness uh but doesn't really say anything specifically

about it and so it seems to me

awesome

Ali want to add anything on that

uh well yeah as far as I know sentience um inactive in principle literature is used uh in a rather

technical sense in which um sentient beings are just uh are they're they're the beings that are

responsive uh to the external stimuli so that's as simple as it gets uh and uh because of its simplicity

uh it uh encompasses a whole array of things as being sentient and uh it's not necessarily uh restricted

to biological sentience or uh I don't know uh any thing akin to uh to the level of human intelligence

for human sentience and so on so uh it can basically be applied to uh include artificial sentience and uh everything even uh virtual sentience such as I don't know the kind of thing that's just been implemented algorithmically in the software and stuff so uh as it it was a subject of a rather repeated debate a few months ago uh I mean the meaning of the sentience is used in active inference literature but in all honesty it's not a novel use either because uh this the same very definition of uh sentience that's used in active inference is almost exactly the same as uh one of the definitions uh from Oxford English dictionaries so uh it's not as controversial as it seems but about the other uh terms such as awareness and consciousness well uh I don't believe we have any sort of consensus around

those words yet in active inference literature although uh in recent papers around um um or mum model of active inference uh Maxwell and others are trying to converge um a consensus um on the word on the notion of

consciousness in terms of active inference

um

yeah what great topics and also how cool with chapter 10 published about a year and a half ago maybe written in the previous months leading up to one and a half years ago how how there are certain similarities and continuations of the discussion um

yeah

Neil I really like the um

the point about having us how how um

active inference how it stays it's unopinionated to the extent that it's composable

as a describing system

those properties are kind of like convex concave

and there's a lot to say probably about why certain uh expectations are around

certain kinds of x of accounts scientific accounts or just certain kinds of rhetorical accounts like

what why that why some are some way and some are not that's the that's the active inference for the

social sciences um but that's a whole thing

having the descriptive framework

cartography

compositional cognitive cartography

as per

the category theory work and and that whole area especially coming into play and the mechanics and the dynamics

it's like

newtonian mechanics whatever you say about it within the system it doesn't have a preference for something moving

left or right or

up or down or something like that
that's just a bizarre
feature to bake into a measurement framework
so

when we look at act implicit really is presented in this textbook
whether as a behavioral researcher like the figure 9.2
metabasean
so whether we approach it from this kind of like classical behavioral researcher or just researchers
perspective
perspective
or we approach it from the constructive perspective
with uh chapter six and and all of that
from six or from nine
uh science and engineering

science and engineering
eventually
maybe
uh it'll come to be seen as an account
of the cartography
so

hence that that divergence with how the the book lands of course this isn't a policy
politics

it's not to say that there's like a perfect text or anything like that
but i think that's kind of the area

i know it's a bunch of random thoughts but it's chapter 10.
these are the funny things about the book that we only really
come into

relation with when we really
stay in the game and and compare how it actually was
so we can look at some specific parts of 10
with the kinds of perspectives that that people
um mobilize today
based upon like for example only the last few days
um

different experimental studies
i'll bring in one link

just as an example of
of course only only in for educational use
this linkedin post
pristen's ai law is proven
fvp explains how neurons learn
anonymized to protect the innocent
um

pristen's
or everyone's slash no one's
ai
law
proven
is
is

explains how
what

and if you think about this in light of of how we've been thinking about 10 then and now
how does this new artifact
from
today
seem to you
yes
ali

well actually i haven't uh read the article in its entirety but uh from uh what i've seen uh
it's not fundamentally different from uh the previous work done in establishing a kind of empirical basis for
active inference but i'm not sure why it gained so much attention now because uh to my mind it's just a
continuation of the previous work done by the same group of authors uh establishing uh
these kinds of these kinds of these kinds of these kinds of these kinds of empirical basis for active inference
but uh i can understand that it can seem as the kind of culmination of this trajectory of this line of research but
uh i'm i'm not sure if the

if there's something fundamentally new or i don't know unprecedented explored in this recent paper
or it's uh it's a kind of um i mean as i said a culmination of all the previous uh efforts done
uh in this area uh but on the other hand uh of course because it's published in nature
uh it would attract lots of media attention and um as we we've seen in the past few days on Twitter or i
don't know X or anything uh uh it's the kind of uh you know um i i'm not saying it's kind of
um uh it's reached the level of hyperbole or something but uh
again i'm not sure if it's quite beneficiary for active inference literature if these kinds of

um I mean this kind of news gets out of proportion and delivers I mean it it wouldn't be able to deliver as much as it promises uh so as a case in point that Denise Holt uh image here seems to me a little bit out of proportion I don't know what you think but um maybe it's just my personal opinion that it's always better to be uh uh to to keep some restraint about these things and uh let let everything loose and uh uh because uh it has it can have dire consequences for any future research if uh um it it it it doesn't get uh what it uh promises or it doesn't deliver what it promises or something like that so that's just my personal opinion on that

excellent Ali very measured very diplomatic Neil yeah I'm I'm only just looking at the article now um yeah you got anything to add

I'll just bring in one other um cultural artifact that that like uh back in the game I just remember being like again everything everyone's base optimal so it's it's I'm not saying I would have done anything differently I'm not saying anything like that I just want to point to these um historical um image meme and everything this was the year that the alias interview with Martin Fortier and Carl had been um well all you need is the title but what else a funny thing I just thought okay I know that it's the article is not ever defined by the headline etc etc genius neuroscientist so okay speculative but singular and then the generality connected to the requirement to understand a specific mind which I think is the same um cognitive associativity between like Nate that's the whole Markov blanket Laplace approximation Natoni mechanics like that we always talk about so that's my first take is just like okay well yeah he may work here or there but um also Stanford doesn't say that it's Gordon's law of collective behavior but also there's a lot of different norms around communication and another adjacency to kind of explore to based upon how we're talking about active like to say the least as at least instrumental and realist but also as I again um all of you have brought up in previous discussions like there's so many other uh perspectives interpretation so if a statistical paper came out some gene expression study from 1992 or some uh height blood pressure risk related study Pearson's regression law is proven linear regression explains how something happens that's also playing into this kind of like scientific positivism again not necessarily the place for the whole falsificationism of science disproves doesn't prove but to to see here's the um what does science do how is science related to law whose ideas are whom's what acronyms do you already have to know an account of a natural process legitimizing um or endorsing even applications of some similar technical method in a different setting a lot of rich image meme and and uh media fun ahead of us

yeah and just throwing in memes the Ik99 i don't even know but what if uh someone's like this is a uh you know room temperature super cognitive process actually Ik99 it was exactly what i had in mind when i talked about uh yeah promising something that can be delivered yeah i again there's so there's so much more to say on this like it's just the complex consequences of different regimes of attention it's so so much more complex and like good or bad or for whom or over

what time scale and uh surfing the hundred foot wave with popularization and also connecting people like just to be explicit to come to the textbook group and to volunteer be an intern at the institute then there's an actionable path from going from like being enthralled by a podcast to working in that that tradition

so

it's all happening in the ecosystem

and we're just

seeing it how we see it from the

a few of us who are here

just in this slash any given moment

all right neil which one of your questions do you think is most relevant

uh well perhaps perhaps the most controversial one is the how important is the theory of free energy minimization so we could do that oh yeah really this is a question of qualitative versus quantitative the notion of free energy is very quantitative but is it

uh how essential is it uh it's a way of formalizing and computing something so so useful from an engineering perspective but isn't it just enough to say what the brain is it's uh a load of neurons in the brains of working in a uh a self-organizing fashion with uh some what to use feynman's phrase ratchet and pull mechanism to to always be or to be moving gradually towards a better solution

um so then then we have this this uh quantity free energy um that's in computable effectively uh that's but we can say well that's that's a macroscopic um quantity but it's it's it doesn't get used anything in any way um as as it does in some machine learning algorithms it's uh so i think i say it's purely epiphenomenal it doesn't it doesn't actually do anything other than well to help understand what's going on um so how how important is you know that whole uh quantitative side of things to the to all of the uh the qualitative aspects of active influence

um implicit is is my my thought that it isn't uh it isn't essential

yeah this is a very interesting question it's a very jf claudia type point

like how different can it be than outside in variational bays contrasted with jf's constructed lego symbolic implementation with no no equations no variational updating so i think that's a really important area to explore ali go for it

um well actually um looking at active inference framework um i mean um superficially we can uh we can see lots of similarities and even um i don't know probably some uh identical uh conceptual components uh with some other frameworks such as predictive processing and uh predictive coding and so on uh but in my opinion the main difference of active inference uh one of the main difference of active inference from other related uh frameworks is exactly what fep enables active inference to do namely uh it puts uh the optimization of surprise though as the main character of the whole scenario so you see this is uh the novel perspective of active inference as compared to all the other frameworks because of course in all the other perspectives we have uh the notions of i don't know anticipating actions and so one and uh basing some decision making uh to basing the the decision making on the prediction and

anticipate the anticipation of actions and uh also perception uh but uh i mean the way fep reframes this whole scenario and as i said puts the surprise optimization uh uh to the forefront as the main character uh makes it much more elegant on one hand and on the other uh it makes it much more broader than all the other

frameworks because we know that predictive coding and predictive processing uh they're mainly restricted to biological intelligence or biological i don't know cognition and whatever but uh we know from uh at least 2019 that the same mathematical technology can be applied to a whole range a whole array of um ontological

entities and uh it's not fundamentally restricted to uh biological sentience or intelligence so uh uh this is uh the key reframing and the key novel perspective that fep provides uh to active imprints and uh if if you look at it i mean these kinds of um fundamental reframings uh and uh somehow pivoting the whole uh structure of a scientific theory based on just a single reframing of a of an essential component is something that uh that can be witnessed throughout the whole history of science time and time again and some of the most profound insights of um um that's been gained in throughout the entire um history of modern science uh has been uh they they've been gained from uh this exact mechanism of changing the perspective uh based on uh some redefinition and some reconceptualization of uh the established facts and established theories so in that sense active inference uh and its um underlying fep uh to my opinion can be seen as i mean it's comparable to all of these uh code unquote revolutionary scientific theories uh in that regard

probably thank you great chapter 10 ism

super super super well said i think this helps differentiate us again while we're in chapter 10 which is essentially doing this exact work wrapping up and connecting non-technically without figures just connecting the dots 10.3 it also connects us to the cybernetics

side it's non-contentious that cybernetic systems can be modeled input output

it's also non-contentious to do multi-agent simulation

and then coming from whether just a pure pragmatic perspective in a finite world or increasingly from the holographic approach and the chris fields physics as information processing physics of boundaries

information processing and thermodynamics and boundaries also kind of converging on a multi-agent thermo informational input output framework and when we're talking pedagogically

sometimes it could be like well it's like input output but you could have like just an oscillator

like just a sine wave or something like that now also i'm sure the sine wave can be understood

as doing the path lease action all of this that is the generality of the framework and it really does pivot it or have a fulcrum as you described ali as the actual centering of surprisal in principle

in literally in principle and in fact slash practice slash deed the tractable bounded probably approximately correct bounded rationality in this situation variational bayesian mechanics

but quantum hardware software could be used surely other kinds of computational computations can be used but the centering of surprisal in principle with the bounded approximation of surprisal with all of the work of the bounded surprisal being bounded surprisal about a given model can't talk like generally

about the l2 norm for some cohort of linear aggressions you haven't named yet that's just not enough information to say further so the theory takes an essential normative focus on surprisal minimization and therefore bridging to the practice with a tractable tractable uh tractable optimizable bound so no contention with other cybernetics frameworks they want to explain observables as a loss function that's awesome perceptual control theory rl they want to have structured or unstructured or fixed or dynamic cognitive architectures also all good um they want to include action and sense making and use a variational auto encoder or a variational bayesian modeling imperative specifically that's an interesting gray zone question is that implicit fep if it's literally already providing a particular partitioned example that would be very interesting dark fep literature search but it's actually those works which arguably are used every day in terms of statistics algorithms by reframing the scenario it's like hey we don't have to reduce uncertainty slash bound uncertainty slash maximize model evidence on the reward function turn the surprisal bounding on itself purely statistically that doesn't mean it's going to be perfect because first there's the information computation bounds chrisfields uh wolfram all of that and then just practically your implementation is going to be like just realistically put together by you and however you actually put it together so you're talking about in principle like exploring blueprints for buildings in principle without even wondering if they can be built or anything like that or are we really in the game talking about motor disorders in this context of this situation very very very different waters fep and act inf though it can be uh assured as far as we believe that there is a fundamental relationship that seemingly we learn more about brings in more pieces because it's a pretty common useful as well as converged upon forest utility already enabling a lot of the shorter term work to be scholarly and of course all pending etc etc etc but that's as as ali has pointed out as they call the structure of scientific revolutions and so in that way that's an interesting and provocative uh area to explore ali um also on a related point and uh getting back to uh the previous discussion we had um about uh the nature of consciousness sentience awareness and uh so on uh so uh philip paul has this recent book uh called the book of minds uh which is a 500 page tome uh but uh he recently gave a talk to the royal institution um it's titled uh let me see yeah it's what is a mind uh so in this talk uh he he concentrates on just one uh chapter of this whole book uh namely uh the space of uh sorry at the the space of possible minds so more often than not we think of all of these uh concepts related to minds such as intelligence agency uh sentience and so on uh as a kind of one-dimensional spectrum one-dimensional continuum but uh philip paul here the shows that uh it would probably be more um i mean uh conducive to further uh development of the

development of further uh future theories if we conceive a kind of multi-dimensional space of possible minds which can have uh the access such as agency experience intelligence and consciousness and

it uh it draws a kind of multi-dimensional uh space such as this and uh i was thinking active inference can actually provide sufficient tools to traverse this whole uh space of possible minds uh if one tries to put all of these different uh axes uh related to one of the uh key trade-offs that's been uh that unfolds through uh through uh the central equations of active inference so basically uh i was thinking that probably or perhaps the central equations of active inference or can be used uh as a kind of uh mathematical tool to traverse this whole uh possible uh by using uh all the all the trade-offs uh that we've seen uh that we've seen uh throughout the whole book uh such as explore exploit phenomena or

other kinds of dilemmas that we've seen before so uh yeah i just uh wanted to uh share this connection with you because i believe there can be some potential uh interesting uh connections between these two theories

that's great another kind of recent uh connection i was just listening to a little of this machine learning street talk with stephen wolfram and he starts talking about yes the ruliad and and all the other work and it's really also impressive how much he condenses it and how much they progressed and then he's like talking about a physics of the observer and a framework for observation and i was just like that's where we need to kind of like direct time stamps clear ontologically structured rhetoric i mean we're in the farmers market phase still overall and how far away is it really from processing and analyzing on one on one branch processing and analyzing speech to develop appropriate paths and wayfinding and guiding and all of that even if it's stephen wolfram giving him the most informative on ramps just like we would for anyone on the computational side and also to neil's point and focus with what's the qualitative side and the true cognitive the phenomenological side the social side and i think that's one reason why um with with scholarly work like that of mel andrews and and mal and others that integrates discussion around model building it also draws upon the pragmatic and the abductive and the semiotic and a lot of other strongly process oriented we talk about process ontologies um all these different

different features uh being woven in at the very least compatibly but compatibly which is to say open-endedly articulated in principle

that's

of course we're just having discussions who knows what's over under confident just just saying how it strikes me at this moment we wouldn't really want a better imperative based upon what we understand at least what i personally understand than something along those lines that sets the stage for multi-perspectival modeling of diverse intelligences

setting that composability it's like if it doesn't have composability

and the unopinionated uh kind of nature collab as the collaborative project

project that's what we want collaborative project essential project critical infrastructure specific
project ryan smith used a stomach uh electrode to test this hypothesis about the vagus nerve and the
role of this like neuropeptide and this muscle uh pattern then we're we're in the standard
scientific scientific expanded scientific space in the process theory space chapter six chapter nine
did you look at the whole system did you consider systematic factors what what trade-offs did you make
as modeling is the process and pipeline reproducible do we have the code like in chapter nine can you make
a figure about the experiment just like connect it to the figures in the textbook like reviewer type
uh functions but as we often bring up reviewers are not going into whether l2 norm or l1 norm was the
statistically appropriate choice i know in ethology many times we just said we looked at this other paper
or just this is a commonly used approach for this statistical technique
uh
so
putting a lot of weight and focus on the last mile
and the specifics which
it's obvious
like pearson's not brought out every time somebody does or doesn't use an r squared to mobilize an
argument
or make a decision that's so many levels away einstein's not brought out except in jest when the planes are
late so
a lot of lot of um morphological and behavioral development
um
but one of the key practices
um
a one thing that wolfram was saying was like there are certain things about this or that that have to be the
case like he was talking about our continuous time perception
even though it's the hypergraph like the discretized hypergraph
at the lower level
uh with that kind of like averages over all the computations i believe
um something like that
um and then like we're orders magnitude larger and slower we average out we coarse grain talked about that
than the theory of the observer
and he's like yeah there might be like other things about communication or about observation or systems
that also like must be the case
so i think there's kind of like in terms of the anthropic principle slash anthropic fallacy um
it's almost like
i believe that we will through chapter 10
and beyond like a minute hey how many careers
like this is not just hot takes on a podcast

these are many many careers in many many directions of many many academic departments
a lot of research programs startups a lot of funding cycles around what is or isn't plausible
ai winter
all the implications causes and consequences of cybernetics research
it's like this is the umbrella that
is on the table
connected only to other accounts
so barring things that we learn on a week by week basis
so at least scale the pragmatic value
to how relevant it is
and then calibrate your epistemic scalar
to
what and how you do and don't know
like we all do and don't know and only speak from our own experience
we can

have a cognitive modeling analysis
what happens when somebody who is not to pick on any field or any topic or anything like that
someone who's like oh i love that citation
or like that's my job
or like i know that really well or my dissertation was in that what happens
in the moment of um contextualization
not to get too meta but again whether it's artisanal farmers market status
and just like well anecdotally people who are um doing psychiatric work or who are interested in like
for example chronic pain as like several of our um very committed colleagues are
like they prefer this
that can be approached by kind of naively or expertly qualitatively and or quantitatively
so we're in in the end of this um cohort the of uh three of the textbook group
so what is it really about the textbook group
how do we really want to think about its role in the ecosystem
like impact on people's learning
production of different materials for different
consumptions and publishings
what do any of you think about that
um

just to give one other random take just kind of unstructured thinking about it is like um
um i was thinking of walking down the aisle at a bookstore and there's different frameworks

catching or vying for attention and i know it sounds obvious that it has to be differentiated
but pointing out that there's differentiation
on multiple axes
as a consequence
of
it sounds wild or weird or however you want to see it at first but the minimization of surprise
is as a central phenomena
and that
simple starting positionality or stance
along with some other core memes that i think needs to be worked out and understood the relationship
to like embody and all this other stuff
that rolls downhill differently
helps you rethink differently
see a lot of phenomena
maybe theoretical transdisciplinary synthesis is not your
um trajectory in life that's totally chill
obvious it doesn't need to be said but just to be clear
that this is
in that
just the where and the when and the how and the who
in relationship to
the
um
actual
which we saw
uh
at least the the movie poster for
earlier in our discussion
uh
and also one other point is
you see
given the rapid development of active inference literature since the publication of this book
i believe
it might not be a terribly bad idea if
beginning from the next cohort
uh
uh
uh

we somehow begin to
incorporate more
uh actively
some of the
findings of
recent literature into the discussions because
i mean
with this
the
pace of development that active inference is experiencing right now
uh
uh
it might sound a bit grandiose but i
think it's not
unlike
the situation
that quantum physics experienced
in the early 20th century
so
as we know
from the history of quantum physics
uh
uh
almost every year and even
uh
every few months
radically new
uh
understanding of quantum theory
uh
appeared which
fundamentally shaped
what
would be now as
quantum physics
and i believe active inference
kind of
taken a similar

trajectory
because
with
of course not
all of the
um
papers that's been published
since the publication of this book
but some of the
uh
groundbreaking papers that really
shifted our understanding of what active inference is all about
and uh
somehow refined
the more vague concepts
that we still see in this book
and try to make them more precise and
make it more coherent with the rest of the
theory
so
yeah
as i said
in my opinion it might not be a bad idea to
look at some of these
um
developments
from the next beginning from the next cohort
thanks a lot
Ali
Neil
yeah
well as
as the discipline grows
with any
new area it grows it gets beyond the point where
where any individual can keep track of things
um
um

i was wondering i mean
well maybe the active inference institute is
effectively doing this already but i'm just imagining
if if there was an annual report let's say the institute
produced an annual report
providing an overview of of what happened
the significant developments
uh in in the last year

um
so that's you were talking earlier about
there's so many different ways in
that it touches on

uh
that you know the people might

um
well have some involvement to do with active
inference but they only know this this tiny area

uh
how would you get them to have some
bigger perspective
where there's there's education such as this book
but how would you enable them to carry on

um

um

um
maintaining that
because
because they can't be reading thousands of journal papers

uh
a year
just on the active inference when
that might actually only be a relatively small part of their
their wider research

and

um

whereas the institute should be able to provide some
oath

is it it's not i'm not just thinking a review of um
some say said this and so said that but some
sort of bigger perspective on on where
on what's happening so like a whole load of new articles
new publications were in
in this direction say social sciences and
and that to to help people

thanks these are such good comments i really agree
ali and i and now actually kind of like pseudo-templated
a little bit like a research roundup
so it's a seed project that's just one possible mechanism not to not jump to a
solution or anything like that but like

i also totally agree like yeah we fixed some
errata and typos we aggregated some people's input
that's
that's local refinement and it's adding value and so on but
then

okay now jonathan and ali have pdfs on this equation that equation
that's like

and then you have

sanjeev's work

as all of you

know

and uh

the annotation work is infinite and i think we've in some ways laid out what certain paths look like
like for example using the ontology seriously

annotating equations with the verbal descriptions

getting all the figures boxes tables eventually even paragraphs into tabular form

like we have the massively collaborative possibilities and that's not to say they're even being underused at
all

by human and or non-human agents

it's just part of a co-creation

so it's not about just like um just like single dimensional perspectives on consciousness or anything like that
uh

are are limited if

if if not worse

um

multiple projects should speak to the bigger perspective

i think there's the important observational role

like here's how surprise was used in this paper at this time

i associate that with the kind of trust finder architecture like rhetorical search assertion not to get too technical but

different expressions are made in the ecosystem

ingressed and or intentionally

and

and

we have different strategies and approaches

and projects

that are anticipatory and maintenance oriented and historical

and

to improve our sense making

there are some interns who are doing such

amazing work

and that's only the fraction

some fraction of people

um

who are not just in their own way and in their own game which is everyone

so like

what we're

what we can really do

i'm i'm happy to have these fundamental conversations

i i just i

we're we're not gonna be able to enumerate or evaluate or anything like that but

par something you know keeping the par flame alive

maybe with a discussion changing the time ollie and i will have completed our background videos

and then we can kind of

assess like here's the roles

here's the first versions of the background materials

go through it

at these you know small groups so there could be a lot of decentralized ways to continue that or at

least just low preparation ways for different facilitators

and then

maybe we need cohort two of chris fields

maybe we need a special integration

units and projects

like that just having a reading group follow a while in social sciences

and follow Chris in in the you're you're going to be intersecting and and

uh engaging and still on the fractal wave

and we're always going to be only using the systems that we have

can't be waiting for future tools but we will use them as they arrive

and just supporting the sustainability of that for the ecosystem is our focus

yeah

another thought just as we're in kind of last minutes of 10. so it's like all this theoretical scope

yeah it could just be another um node in the same space but the claim is kind of like it generalizes

a lot of them relates a lot of them formally um maybe even envelops all of them um

at least socio uh socio technically if not even way way way deeper and broader and and everything

like that so it's like however much theory scope you think this covers maybe you're super familiar

with literature you're for every single citation you're like oh yeah you know exactly why it's there

then this argument is poised to your adjacency because this is the perspective of the authors

or if you're not super familiar with these areas it may be sufficient just to understand them

as being like broad topics of interest if not addressing important systems in the real world

but when we really think about these as being the major threads of accounts and statistics for

cognitive behavioral systems it's not like there's some other like secret 11 theory

that's just being like censored in the citation literature i mean i'd leave it to a meta-analysis

to say more but but nobody reads this and says wow you're you're ignoring like this entire massive

thing yeah maybe go into more detail on adaptive resonance theory or go into more detail on this

on one given area but when the broad threads are represented and the arguments are made formally in

text based upon an implicit first principle the scope and the relevance of the learning has to

be this theoretical scope and the consequences on systems of interest doesn't mean that it has to be

overwhelming in the moment but people bring up real things in the textbook group and it is about real

things because education is about real things i would like to hear any of your all's

thoughts on chapter 10 now as we close cohort three

empathize with empathize with empathize with empathize with empathize with empathize with empathize with
not us.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Any thoughts?

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

It's a different system of interest.

It's not the same thing as learning.

Learning epistemic value, that's why it's sometimes fun just to learn about things for their epistemic value.

There's parts that we know has pragmatic value.

There's parts that we know have epistemic value.

There's parts that we know have epistemic value.

What other imperative are you acting and inferring under?

Are you acting and inferring under?

And then one route that Ali highlighted very articulately earlier was the central character, main character energy for surprisal.

And a kind of close in that nexus is the qualitative, but formal, potentially qualitative compositional about existence, existentials.

And that ties it to homeostasis and all of that, ties it to observation and all of that.

It opens up another degree of freedom, the axial constant, Mobius strip.

It's not about da-da-da, it's about how you got there.

Like, this describes systems with persistence.

Do you have a better descriptor?

But that doesn't mean that constructing it is a guarantee of future persistence.

You know, past returns does not guarantee future returns.

Or whatever they say there.

But you may be able to say certain things, but the existentialism is a natural fit in information landscapes

because it kind of blurs.

And it doesn't actually blur, but it conceptually integrates the measurement and the existential on a kind of common footing.

Super general analytical representations, the exact same kind from like variational physics mechanics, information theory, quantum, category theory, dynamic systems.

All their notation, special case.

All the math that people are using to describe systems of interest.

Compensationality.

Recognition of new underlying special case phenomena.

Just as Buckminster Fuller always pointed to.

Recognition of special case phenomena.

Upon generalization.

Also related to the revolution model that Ali mentioned.

And that is the thing.

You just have a particular partition.

But then.

You have them fitting an L2 norm.

Which is not unlike what a structural equation model is actually.

And structural equation modeling is used in SPM.

And I'm sure there's all kinds of different variants and stuff, but it's like, that's like a kind of massively linear parametric.

Can include time series, just like all the Sarima methods and all of that.

And a lot of the same ability to describe like changing causal relationships.

Little known fact existed in SPM.

So a lot of the practice.

In principle.

Was totally, totally addressed by SPM textbook.

Various modules became increasingly recognized centrality of action, the pragmatic turn in neuroscience, all that.

Those.

Those.

Recognitions of were either articulations that were not understood to have already existed.

Like the idea that you could just extend.

The kind of Coleman filter like.

Predictive processing algorithm arising in video compression.

And then just like.

Pop action in.

And get active inference.

With all of these even immediately apparent philosophical implications for people who are studying those topics.

And the maintenance of the key feature with the physics based first principles singular imperative.

So recognitions of articulations that that suggested.

Also profitable and fruitful directions important critiques and and re understandings.

And developments.

Not just related to notation.

The.

Generative model and generative process, which as we've discussed more recently.

Is not the way that it's often being discussed.

So does that lead to an entire secondary discussion?

Or do we just continue forward?

Mostly with.

Such and such.

But the particular partition.

But the particular partition.

And the.

Over the last years.

The refinement of so many different aspects such that we're basically.

Like preparing to scramble.

Not really.

How to even synthesize.

Hence the ultra descriptive deflationary ontology.

And assertion based architecture that we hope is going to bring in a lot of.

Diverse ways of knowing and remodeling and sense making.

And.

And.

There will be a lot of different.

Angles and overlaps.

Like what if there's a statistics professor who's just feels a certain way about.

This.

How does that situate them in relationship to the theoretical apparatus.

As understood by any other given person.

Or by any other group of other people.

So how to really retain the.

The.

The.

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It's comforting to.

Expect and prefer that we have the framework that is on the path towards that.

Composability extensibility, et cetera, et cetera, all the phenomena that we discuss.

And what's the other.

Aspect to it.

and the textbook's not life

so how to emphasize that too

and really integrate it in people living and learning

make it what it can be

so we could definitely think a lot and uh no specific dates for

cohorts are set but maybe you know who knows social sciences cohort two

chris fields cohort two

par

it's people's attention slash nestmate level contributions that

actually embody those things and so after the acclimation of like the kind of woe

there's still so much work and it will continue to be

so

it's been a total

honor to neil ali and esmail and everybody else who contributed totally to this cohort

um any any last comments on cohort three

fascinating journey as always

so

the summary well summary chapter will always do that sort of throwing everything there is in front of you

but yeah there is so much to the whole to the subject isn't there i've brought in from so many different places

it really is

it really is

yeah it's been great listening to you

yeah we've all

we've all learned a lot it's been a very great end so i guess next week the the day of the symposium i mean

we can discuss more but we'll see what we may want to do or how we want to proceed in the coming um cohorts so thanks again see you all next time bye

thanks everyone thanks

bye

thanks